

BIRSA MUNDA, India

WMO: 427010

Lat: 23.314N

Lon: 85.322E

Elev: 655

StdP: 93.70

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.5	9.7	-1.8	3.5	23.6	0.0	4.1	21.8	5.4	21.4	5.0	21.3	0.4	320	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.9	39.6	20.3	38.2	20.4	36.9	20.4	26.2	30.5	25.7	29.8	25.3	29.1	3.4	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	22.0	27.8	24.8	21.5	27.4	24.4	21.0	27.0	86.2	30.6	84.3	30.3	82.6	29.3	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.8	5.0	4.3	6.3	41.1	1.0	1.3	5.5	42.1	4.9	42.8	4.4	43.6	3.6	44.5		
			4.5	27.9	1.0	1.1	3.8	28.7	3.2	29.3	2.6	30.0	1.8	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	16.9	20.2	24.7	29.3	30.6	29.1	26.4	26.3	26.1	24.2	20.5	17.3	(6)
(7)		DBStd	4.90	2.04	2.60	2.64	2.50	2.40	2.85	1.41	1.22	1.37	1.78	1.86	1.74	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	109	55	10	0	0	0	0	0	0	0	0	3	42	(9)
(10)		CDD10.0	5222	213	287	456	579	637	574	508	504	482	439	316	226	(10)
(11)		CDD18.3	2290	9	63	198	329	379	324	250	246	232	181	69	9	(11)
(12)		CDH23.3	23247	109	563	2051	4368	5117	3776	1958	1859	1712	1226	427	82	(12)
(13)		CDH26.7	10312	9	143	947	2547	3012	1912	479	426	464	324	47	3	(13)
(14)	Wind	WSAvg	1.6	1.3	1.5	1.8	2.0	2.3	2.1	2.1	1.8	1.6	1.1	1.0	1.0	(14)
(15)	Precipitation	PrecAvg	1370	19	18	21	18	65	242	334	314	246	84	9	7	(15)
(16)		PrecMax	1884	98	63	80	88	138	542	625	562	439	338	58	57	(16)
(17)		PrecMin	870	1	0	0	0	12	66	178	159	36	0	0	0	(17)
(18)		PrecStd	249	23	17	20	22	38	107	108	107	103	87	18	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	32.0	36.8	40.8	40.8	40.6	33.1	32.1	32.3	32.1	29.2	26.9	(19)
(20)		0.4%	MCWB	15.1	17.1	17.7	19.1	20.0	21.0	25.1	25.4	24.9	21.9	19.1	16.1	(20)
(21)		2%	DB	26.0	30.0	34.6	39.1	39.7	39.0	31.6	30.9	31.2	30.7	28.0	25.2	(21)
(22)		2%	MCWB	15.0	16.6	17.9	19.1	20.2	22.3	25.2	25.2	24.6	21.8	18.6	15.5	(22)
(23)		5%	DB	24.4	28.3	33.0	37.8	38.5	37.0	30.3	30.0	30.2	29.5	27.0	24.1	(23)
(24)		5%	MCWB	14.3	16.0	17.4	19.1	20.5	23.0	25.1	25.0	24.4	21.8	18.0	14.8	(24)
(25)		10%	DB	23.0	26.8	31.5	36.2	37.1	34.9	29.2	29.1	29.4	28.5	26.0	23.0	(25)
(26)		10%	MCWB	13.5	15.3	17.2	19.0	20.6	23.7	24.9	24.9	24.3	21.6	17.7	14.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	20.2	21.4	24.7	26.3	26.9	26.5	26.5	26.3	25.2	22.1	18.7	(27)
(28)		0.4%	MCDWB	21.5	25.9	28.7	31.9	32.2	32.2	30.2	30.0	30.0	28.8	25.4	23.2	(28)
(29)		2%	WB	17.0	18.8	20.2	23.3	25.2	26.1	25.9	25.8	25.6	24.6	21.1	17.4	(29)
(30)		2%	MCDWB	21.9	24.9	28.4	30.8	31.6	31.5	29.5	29.1	29.2	28.0	24.2	22.1	(30)
(31)		5%	WB	16.0	17.7	19.4	22.1	24.5	25.6	25.5	25.4	25.2	24.0	20.2	16.4	(31)
(32)		5%	MCDWB	21.2	24.1	28.1	30.5	31.2	30.7	28.7	28.4	28.5	27.3	24.1	21.6	(32)
(33)		10%	WB	15.0	16.8	18.6	21.2	23.9	25.1	25.1	25.1	24.8	23.4	19.3	15.6	(33)
(34)		10%	MCDWB	20.5	23.4	27.6	30.3	30.7	30.0	28.1	27.9	27.8	26.2	23.8	20.9	(34)
(35)	Mean Daily Temperature Range	MDBR	12.9	12.7	12.9	13.1	11.9	8.7	5.4	5.4	6.1	8.8	11.6	12.5	(35)	
(36)		5% DB	MCDDBR	14.4	14.1	14.3	14.3	13.4	11.8	7.3	6.9	7.9	9.9	12.0	13.4	(36)
(37)		5% DB	MCWBR	5.8	4.9	4.3	4.4	4.3	3.7	2.5	2.5	2.8	3.6	4.5	5.2	(37)
(38)		5% WB	MCDDBR	11.2	11.5	12.4	13.2	11.9	8.8	6.1	5.9	6.7	7.2	9.0	11.4	(38)
(39)		5% WB	MCWBR	5.2	5.0	4.7	5.2	4.8	3.4	2.6	2.6	2.9	3.2	4.0	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.452	0.449	0.495	0.535	0.600	0.637	0.561	0.546	0.547	0.524	0.517	0.483	(40)	
(41)		taud	2.000	2.057	1.956	1.896	1.791	1.759	1.958	2.017	1.991	1.967	1.878	1.935	(41)	
(42)		Ebn at Noon	793	833	812	786	732	700	755	768	758	752	723	742	(42)	
(43)		Edh at Noon	160	161	186	201	222	228	187	176	177	174	179	165	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.94	4.84	5.79	6.40	6.24	5.08	4.07	4.21	4.34	4.55	4.10	3.68	(44)	
(45)		RadStd	0.27	0.29	0.24	0.37	0.28	0.46	0.36	0.22	0.32	0.44	0.21	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page